SUPPORTING ADULT GUIDANCE

Direct and inverse proportion

Percentages, Ratio, Growth and Bounds · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Proportion Explorer, learners choose one of four situations: pay by the hour ($y = 12x$), sharing a 24-hour job ($y = 24/x$), a ball rolling down a slope ($d = 5t^2$) and fencing a square field ($P = 4\sqrt{A}$). They step x through values chosen so every output is exact; each is added to a table and plotted on numbered, labelled axes. The table's last column is the test that stays constant ($y \div x$, $x \times y$, $d \div t^2$, $P \div \sqrt{A}$). Showing the curve reveals the formula. Seven questions and a stretch follow (why $y = 24/x$ never meets the axes, with rectangles of area 24 and each axis highlighted in turn). Movement sorts tables by kind; Reflection has true-or-not statements; Creativity asks learners to choose the kind and k so a graph passes through target points.

Driving Question: *When one doubles, what does the other do?*

The mathematics, in plain terms

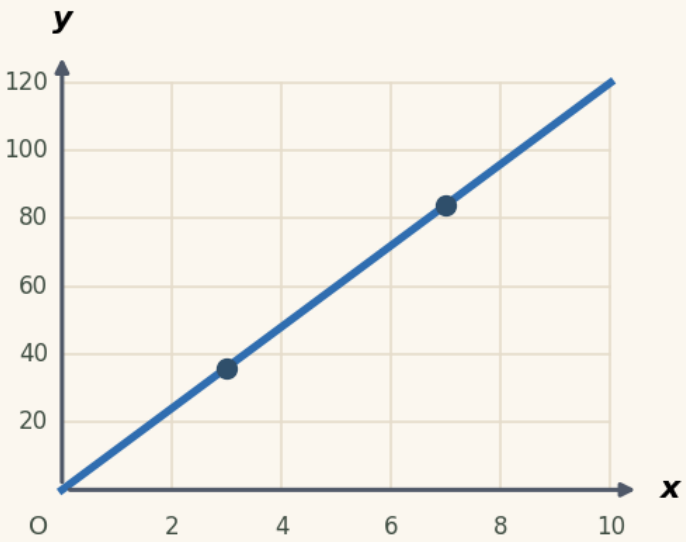
$y \propto x$ means $y = kx$ for a constant $k \neq 0$; $y \propto 1/x$ means $y = k/x$, equivalently $xy = k$. Extensions: $y \propto x$■ and $y \propto 1/x$■, $y \propto \sqrt{x}$. Direct proportion graphs are straight lines through the origin; inverse proportion graphs are rectangular hyperbolas with both axes as asymptotes, since $xy = k \neq 0$ excludes $x = 0$ and $y = 0$. A linear relationship with a non-zero intercept ($y = mx + c$, $c \neq 0$) is not proportional.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Direct proportion: pay by the hour

Pay y (£) is directly proportional to hours worked x. 3 hours pay £36. How much do 7 hours pay?

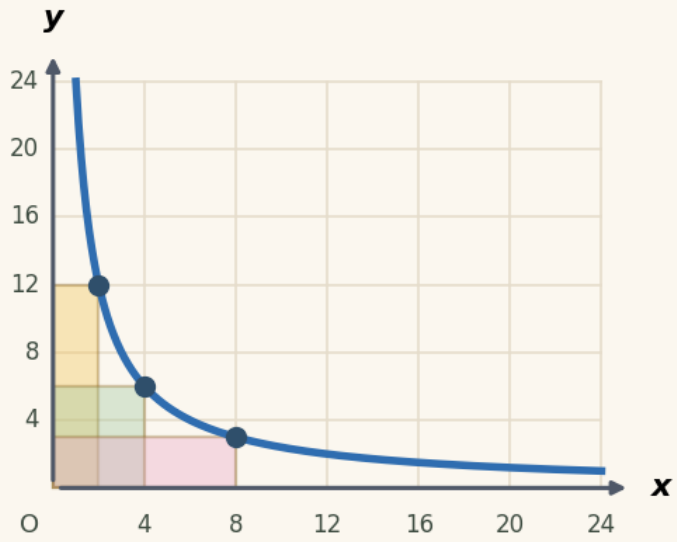


1.	Direct proportion	$y = kx$
2.	Substitute $x = 3$, $y = 36$	$36 = 3k$, so $k = 12$
3.	The formula	$y = 12x$
4.	$x = 7$	$y = 12 \times 7 = 84$

7 hours pay £84. The graph is a straight line through the origin.

Example 2 — Inverse proportion: sharing a job

The time each worker takes, y hours, is inversely proportional to the number of workers, x. 2 workers take 12 hours each. How long do 8 workers take?



1. Inverse proportion $y = \frac{k}{x}$, so $x \times y = k$
2. Substitute $x = 2$, $y = 12$ $k = 2 \times 12 = 24$
3. The formula $y = \frac{24}{x}$
4. $x = 8$ $y = \frac{24}{8} = 3$

8 workers take 3 hours each. Every shaded rectangle has area 24.

Tables and graphs before formulas

Learners see a constant appear in a table and a shape appear on a graph before they write $y = kx$ or $y = k/x$. The formula then summarises what they have seen, and finding k from one pair follows naturally from Lesson 1's “find one, then scale”.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Proportion Explorer: four situations, the constant test and the graphs. Seven questions. Stretch: why $y = 24/x$ never meets the axes.	~16 min
Movement	Spot the proportion: sort tables by kind.	~6 min

Reflection	True or not? Straight lines, halving and squaring.	~8 min
Creativity	Make it fit: choose the kind and k.	~10 min
Rest	More hands, lighter work.	~2 min
Nutrition	Intellectual: prediction from one pair of values.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking any straight line shows direct proportion.
Ask: “When x is 0, what is y? Does the line pass through the origin?”

Treating inverse proportion as “one goes up by the same amount the other goes down”.
Ask: “What happens to y when x doubles? Is $x \times y$ the same each time?”

Adding the change in x to y (35 when $x = 5$, so 39 when $x = 9$).
Ask: “Is it multiplying or adding? What is $y \div x$?”

Using $k = y \div x$ for inverse proportion.
Ask: “Which stays the same here: $y \div x$, or $x \times y$?”

Forgetting to square x in $y = kx^2$.
Ask: “What is x^2 here, before you multiply by k?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the unitary method (Lesson 1), substitution, and squares and square roots. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says that if 2 workers take 12 hours, 8 workers take 48 hours. The aim is a question that brings back the situation.

- “With more workers, should the job take longer or shorter?”

- “How many worker-hours is the job altogether?”
- “So how many hours each, for 8 workers?”
- “Which stays the same: hours ÷ workers, or workers × hours?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Do learners need the symbol $\propto$?
It is introduced once alongside “is proportional to”. Writing the formula with k matters more.

Why include squares and roots?
The specification expects them at Higher tier. The method is identical: write the formula with k, find k, use it.

Is the stretch required?
No. The secure core is recognising the type, finding k and using the formula.

What comes next?
Lesson 3: percentages as multipliers, where a single number again describes a change.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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